

Demo: Ruben's Tube

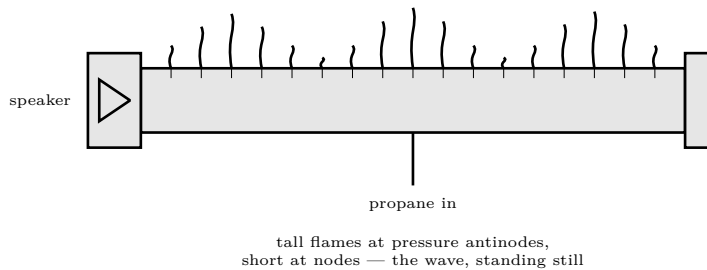
Make a sound wave visible as a wall of flame

The best-looking thing you can build with a drill and a pipe

Lesson Demo · an afternoon to build · adult owns the gas, always

PROPANE IN A PERFORATED PIPE — OUTDOORS ONLY

This is a pipe full of flammable gas with an open flame along its length. Build and run it **outdoors**, never indoors, never in a garage, never under anything. An adult owns the cylinder, the regulator and the valve, and lights it — the same rule as the launcher's fuel. Purge and light promptly: a tube left filling unlit accumulates gas and then lights all at once. Keep the extinguisher on the bench, not in the house.



Predict first

Sound is a travelling wave. Play a steady tone into a closed tube full of gas with a row of small flames along the top. Do the flames ripple along like waves on water, all flicker together, or something else? Draw what you expect before you light it.

What happens	The flames settle into a fixed pattern of tall and short that does not move. Change the frequency and the pattern changes to a different fixed arrangement.
Why	The tone reflects off the sealed end and interferes with itself. At the right frequencies the two waves cancel at fixed points and reinforce at others, producing a standing wave — a wave whose peaks stay put.
Flame height reads pressure	Higher pressure inside pushes more gas out of the holes there, so a tall flame marks an antinode and a short one a node. You are looking at a graph of pressure along the tube, drawn in fire.
Measure the speed of sound	Twice the distance between adjacent nodes is one wavelength. Multiply by the frequency and you have the speed of sound <i>in propane</i> — around 250 m/s, noticeably slower than in air, because the gas is heavier.
The link to the launcher	Same physics, different use: pressure in a closed volume of gas, and gas that burns only where fuel and air are in the right proportion. <i>see the 2. Combustion sheet</i>

Build it

- Pipe.** A 1 m to 2 m length of steel or copper pipe, 50–75 mm across. **Not PVC:** this one has open flame along its whole length, and the pipe gets hot. The launcher's pressure-rated PVC rule does not transfer here — different job, different material.
- Drill the holes.** A row of 1.5 mm holes along the top, evenly spaced every 15–25 mm, all the way down, in a straight line. Mark the line first and carry it the whole length. *see the Craft 1: Measure & Mark sheet*
- Deburr every hole.** A burr changes the flame at that hole and puts a false reading in your pattern. *see the Craft 3: Deburr & Chamfer sheet*
- Seal one end** with a solid cap. The reflection off this end is what makes the standing wave; a leaky end gives a poor

pattern.

5. **Fit the speaker** at the other end over a flexible membrane — a rubber sheet or a balloon stretched and clamped. It must seal the tube while still moving freely.
6. **Gas inlet** in the underside near the middle, with a valve. Adult keeps the valve.
7. **Leak-check with soapy water** before any flame goes near it, at full working pressure. Bubbles anywhere means stop.

Run it

Purge first	Let gas flow long enough to displace the air inside, then light along the row promptly. Do not let it fill unlit.
Find the resonances	Sweep a tone generator slowly from about 50 Hz upward. Most frequencies do nothing interesting; at particular ones the pattern snaps into place. Those are the tube's resonances — exactly the behaviour from <i>see the 8. Resonance sheet</i> in the electricity course, in a different medium.
Count the nodes	Each resonance has one more node than the last. Record frequency against node count; the relationship is a straight line and you can predict the next one before you find it.
Do the arithmetic	Node spacing times two is the wavelength. Wavelength times frequency is the speed of sound in the gas. Compare with 343 m/s in air and explain the difference.
Then play music	Once the measurements are done. A bass line makes the whole tube dance, and it is worth every minute of the build.

Why this is worth building

Standing waves are on every physics syllabus and almost nobody has seen one. A diagram of nodes and antinodes is an abstraction; a metre of fire holding perfectly still while a speaker plays is not. Your boys will be able to point at a node and say what it is, and they will still be able to in ten years.

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